Iterative Method for Modelling the Distribution of Regression Effects in the Cox Proportional Hazards Model

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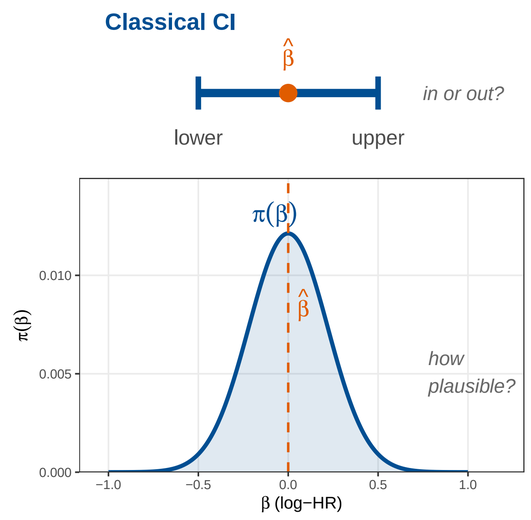
Motivation

Standard Cox inference gives a **point estimate** and a **confidence interval** — a binary in-or-out summary of uncertainty.

Goal: construct a smooth *plausibility distribution* $\pi(\beta)$ over all log-hazard-ratio values — entirely within the **frequentist** framework, without Bayesian priors.



Cox Model — Brief Primer

$$h(t \mid \mathbf{x}) = h_0(t)\exp(\beta_1 x_1 + \cdots + \beta_p x_p)$$

- $h_0(t)$: baseline hazard — unspecified (semi-parametric)
- β_j: log-hazard ratio; e^{β_j}: hazard ratio
- **Wald test:** $Z = (\hat\beta - \beta_0)/\widehat{\mathrm{SE}}$, $p = 2[1 - \Phi(|Z|)]$

Algorithm

Step 1 — Grid. Fit Cox; obtain $\hat\beta$ and $\widehat{\mathrm{SE}}$. Choose m values:

$$\beta_k \in [\hat\beta - 3\widehat{\mathrm{SE}},\ \hat\beta + 3\widehat{\mathrm{SE}}], \quad k = 1, \ldots, m.$$

Step 2 — Bootstrap Wald p-values. Resample rows B times (case resampling):

$$Z_k^{(b)} = \frac{\hat\beta^{(b)} - \beta_k}{\widehat{\mathrm{SE}}^{(b)}}, \qquad p_k^{(b)} = 2\left[1 - \Phi\left(|Z_k^{(b)}|\right)\right]$$

Average: $\bar p_k = \frac{1}{B}\sum_{b=1}^{B} p_k^{(b)}$. High $\bar p_k \Rightarrow \beta_k$ is **plausible**.

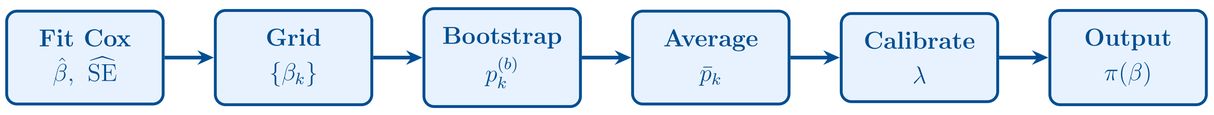
Step 3 — Probability measure.

$$w_k = \exp\left(-\lambda(1 - \bar p_k)\right), \qquad \pi(\beta_k) = \frac{w_k}{\sum_\ell w_\ell}$$

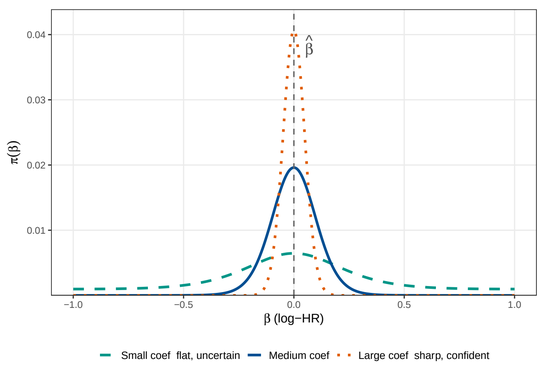
$(1 - \bar p_k)$ measures implausibility; $-\lambda$ reverses the sign so plausible values get high weight. $\lambda > 0$ sets concentration.

Step 4 — Summaries. Mode, mean, and 95% highest-density interval (HDI) of $\pi(\beta)$.

Pipeline



Concentration Coefficient λ



In linear regression: $\lambda = n/\hat\sigma$ (closed form). In the Cox model no $\hat\sigma$ exists $\Rightarrow$ two calibration strategies below.

Two Calibration Strategies for λ

(A) CI Matching *numerical*

Find λ so that the probability mass of $\pi(\beta)$ inside the Wald 95% CI equals the nominal level:

$$\sum_{\{\,k\,:\,\beta_k \in \mathrm{CI}_{95\%}\,\}} \pi(\beta_k) = 0.95.$$

(B) Contrast Ratio *closed-form*

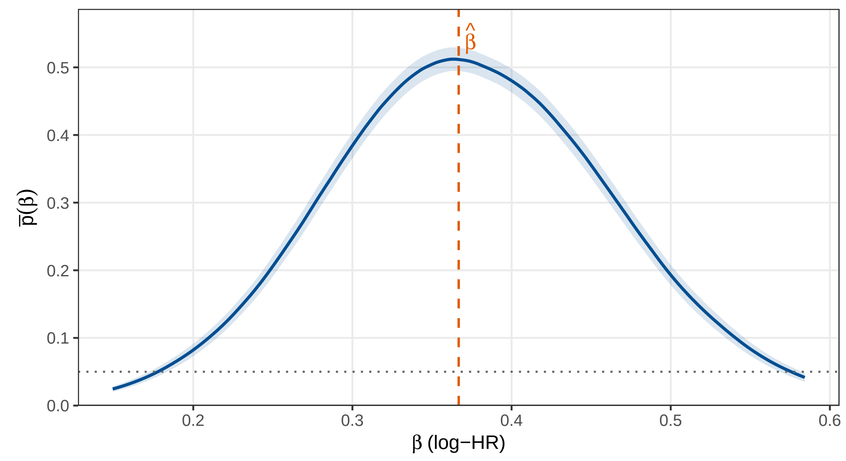
Enforce a fixed plausibility ratio R between $\hat\beta$ and a reference $\tilde\beta_{\mathrm{ref}}$ (median of the grid):

$$\frac{\pi(\hat\beta)}{\pi(\tilde\beta_{\mathrm{ref}})} = R \quad \Rightarrow \quad \lambda = \frac{\log R}{\bar p_{\max} - \bar p_{\mathrm{ref}}}.$$

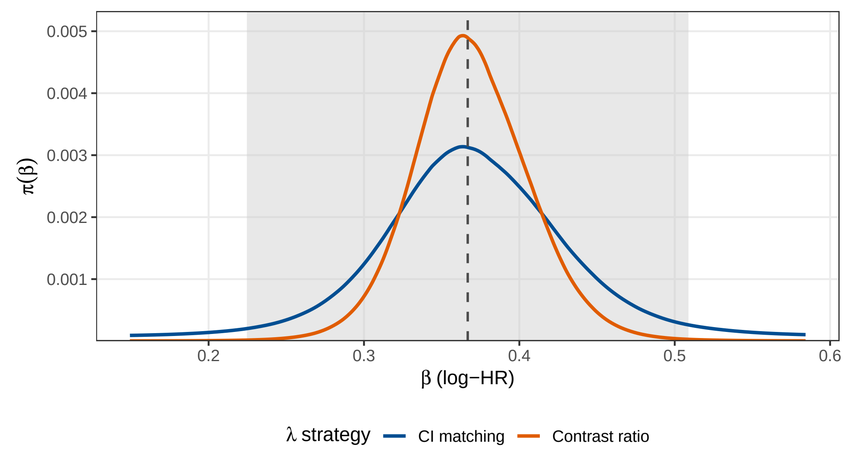
$\hat\beta$ is always $R\times$ more plausible than the reference, independently of sample size.

Illustrative Example (n=300, HR = 1.5, ≈32% censoring, B=1000, m=1000, R = 50)

Bootstrap $\bar p(\beta)$ curve with 95% band:



Plausibility distribution $\pi(\beta)$ — both strategies:



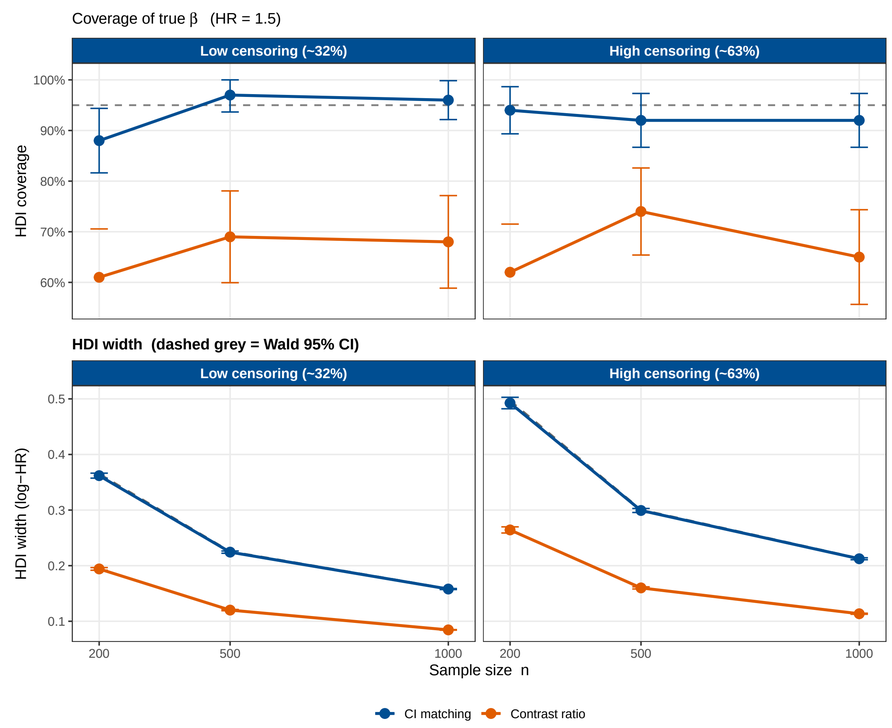
Shaded: Wald 95% CI. Dashed: $\hat\beta$.

Blue (CI matching): exactly 95% mass inside CI by construction.

Orange (contrast ratio): narrower, sharper distribution.

Simulation Study

Design: $n \in \{200, 500, 1000\}$; HR $\in \{1.1, 1.5, 2.0\}$; ≈32% and ≈63% censoring; 100 replicates, B=500, m=500, R=50.



Top: HDI coverage of true β (dashed = 95% nominal). **Bottom:** HDI width vs. Wald CI width (dashed grey).

Key findings:

- **CI matching** achieves near-nominal 95% mass calibration by construction across all scenarios.
- **Contrast ratio** with R = 50 yields consistently *narrower* HDIs — useful when sharper summaries are needed.
- Performance degrades gracefully at small n.
- **CI matching** is data driven method, whereas **Contrast ratio** introduces user defined ratio, which can skew results towards biased initial estimate.

References & Contact

[1] Cox, D.R. (1972). Regression models and life-tables. *JRSS-B*, 34, 187–220.

[2] SEIF, Lubomír; VÍT, Ondřej; HUDÁK, Miriam Helena; ŠTĚPÁNEK, Lubomír. An experimental probabilistic interpretation for confidence intervals in regression. In: LÖSTER, Tomáš; PAVELKA, Tomáš (ed.). The 18th International Days of Statistics and Economics. Slaný: Libuše Macáková, Melandrium, 2024, s. 340–349. ISBN 978-80-87990-34-6.

[3] Seif, L. (2025). Iterative method for testing regression effects [Master's thesis, University of Economics, Prague]. Prague University of Economics and Business. https://vskp.vse.cz/96364_iterativni-metoda-testovani-regresnich-efektu

Supplementary code